

Chapter 1: Uncertainty and Inference

임수빈

Perception = inference about the world

We do not observe the “world state” directly — we observe noisy sensory data.

- Goal: estimate which hypothesis about the world is most likely given the observation.
- This naturally leads to conditional probabilities: $p(\text{world state} \mid \text{observation})$.

Notation used throughout the chapter

Observation (sensory input): A | Hypothesized world state: B

Key quantity: $p(B \mid A)$

Conditional probability in everyday life

$p(B|A)$ = “the probability of B given A”

(A)



p (this book is worth reading |
what I've read so far)

(B)



p (I'll get sick if I eat this apple |
its looks, its smell, the worm...)

(C)



p (that's the mayor's voice |
acoustic information)

(D)



p (I'm going to trip and fall | my
shoelaces are tied in a knot)

(E)



p (switching channels on TV |
the remote has no batteries)

(F)



p (they're the one | personality,
behavior, appearance)

Figure 1.1: Different scenarios with probability judgments. The notation $p(B|A)$ is read “the probability of event B given event A .”

Conditional probabilities are not symmetric

Important: in general, $p(A|B) \neq p(B|A)$.

- Basketball example: most professional players are tall, but most tall people are not pros.
- So $p(\text{tall} | \text{pro})$ is high, while $p(\text{pro} | \text{tall})$ is low.
- Why? Because base rates (how common each event is) differ.

Exam tip: Always check the direction of conditioning and the base rate.

Hypotheses and probability distributions

A “hypothesis” = a candidate world state we consider.

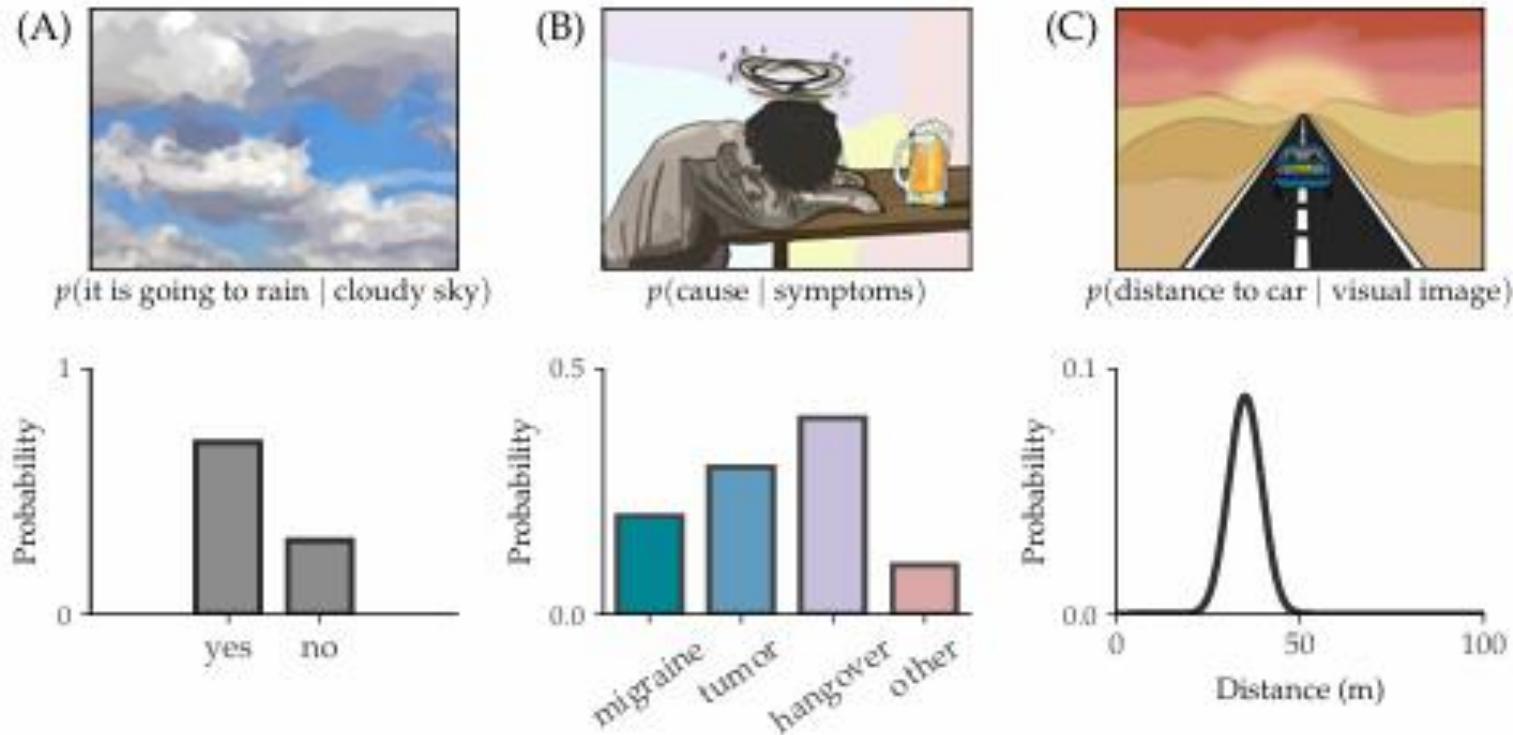


Figure 1.3: Different kinds of probability distributions. (A) Two hypotheses. (B) Multiple hypotheses. (C) Continuous (infinitely many) hypotheses.

Likelihood: how well each hypothesis predicts data

Likelihood function: $p(\text{observation} \mid \text{hypothesized world state})$

Answers: “If this hypothesis were true, how likely is the observation I saw?”

A sharper (more peaked) likelihood means the observation is highly informative.

A flatter likelihood means many hypotheses could explain the same observation.

Prior: what you expect before seeing the data

Prior distribution: $p(\text{world state})$

Represents background knowledge and expectations (learned from past experience).

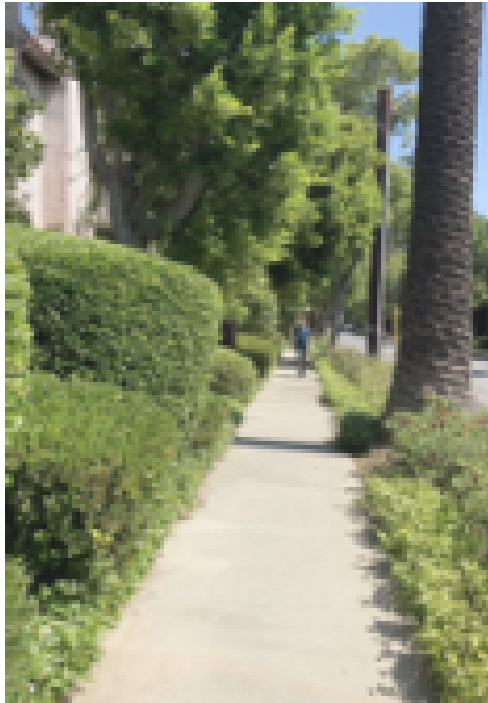
Different people can have different priors in the same situation.

Priors can be “sharp” (strong expectation) or “flat” (weak expectation).

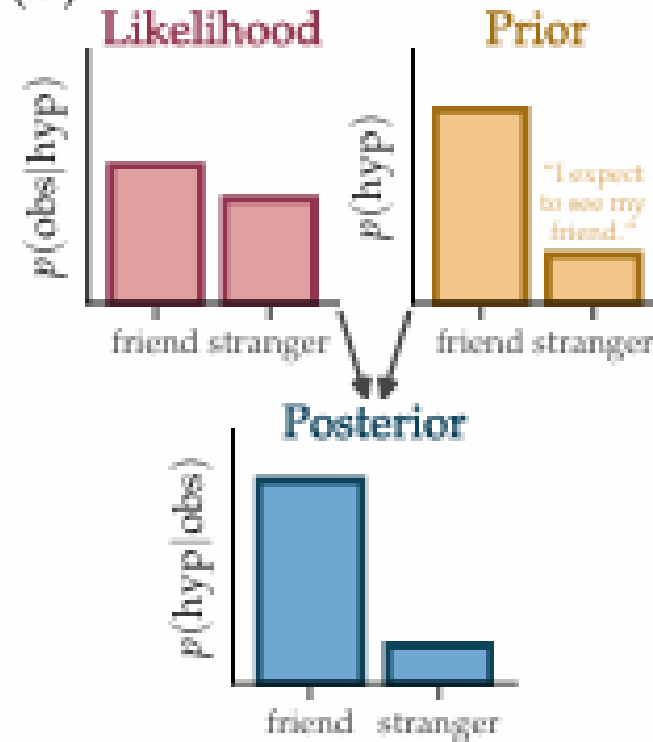
Example: recognizing a friend

Same observation, different priors → different conclusions

(A)



(B)



(C)

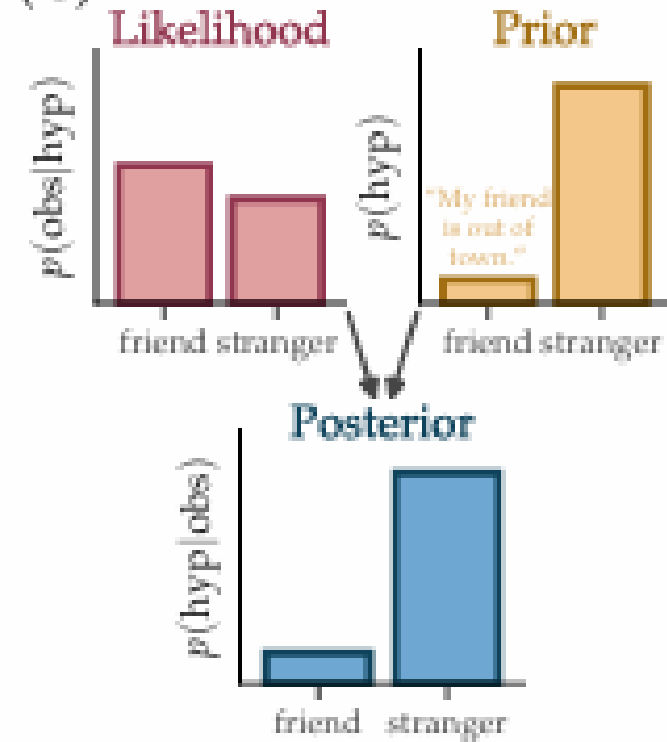


Figure 1.4: Likelihood + Prior → Posterior

Bayes' rule: combining prior and likelihood

Posterior $\propto$ Prior $\times$ Likelihood

$$\textit{Posterior} = \textit{constant} \cdot \textit{Prior} \cdot \textit{Likelihood}$$

Posterior: $p(\text{world state} \mid \text{observation})$ — what you believe after seeing the data.

Constant (normalization): makes probabilities sum/integrate to 1.

If likelihood favors A but prior favors B, the posterior depends on which is stronger.

1.3 Sensory noise and perceptual ambiguity

Sensory noise = stochastic variability in sensory signals (external + internal).

Noise widens the likelihood function → posterior becomes flatter (more ambiguous).

Repeated observations can reduce uncertainty (likelihood sharpens over time).

Key intuition: flatter posterior → more uncertainty

Likelihood widening can happen without noise

Geometric/optical ambiguity, occlusion, unknown illumination...

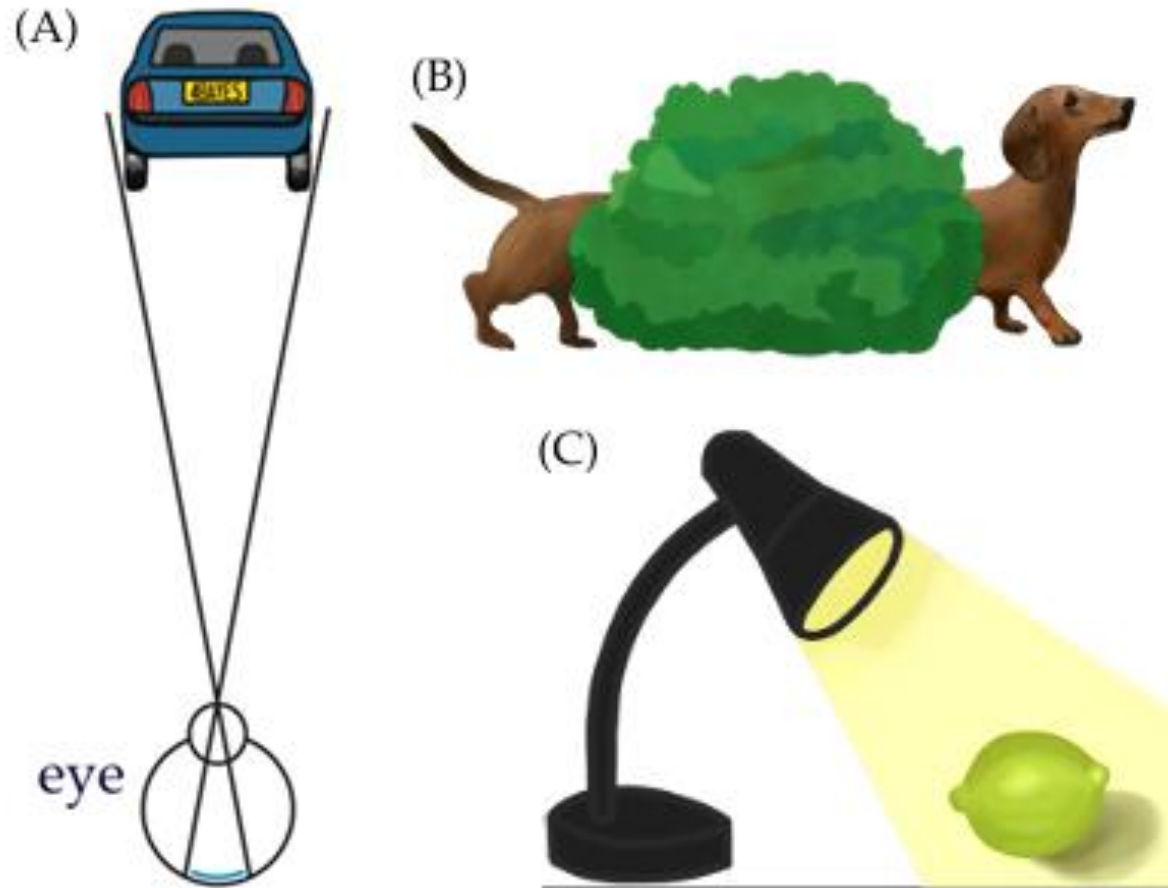


Figure 1.6: size-distance, occlusion, illumination

1.4 Bayesian inference in visual perception: wetness

Shiny floor: wet or dry?

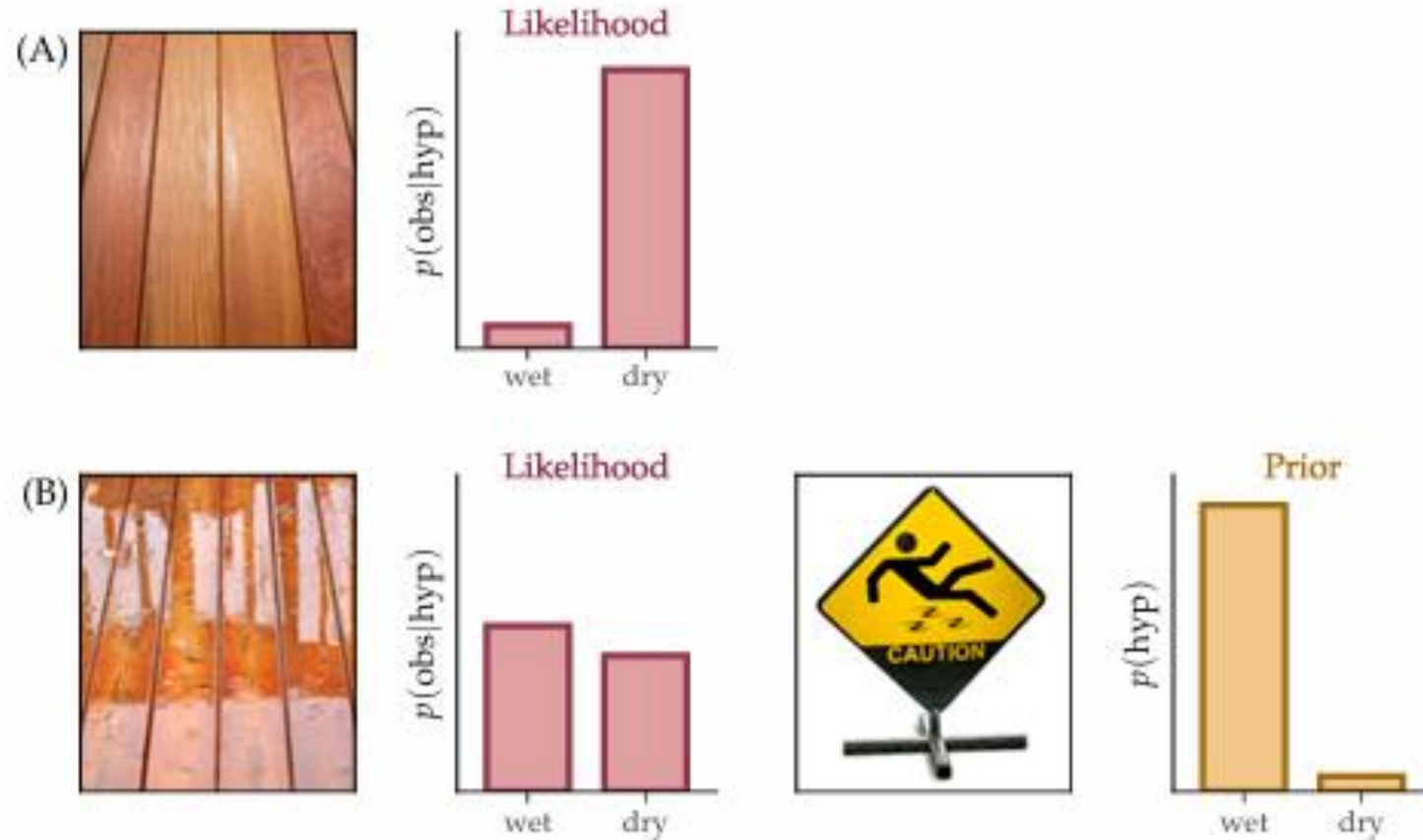


Figure 1.7: likelihood can flip; priors can sharpen

Visual acuity changes the posterior

Same prior, different visual acuity $\rightarrow$ different likelihood $\rightarrow$ different posterior

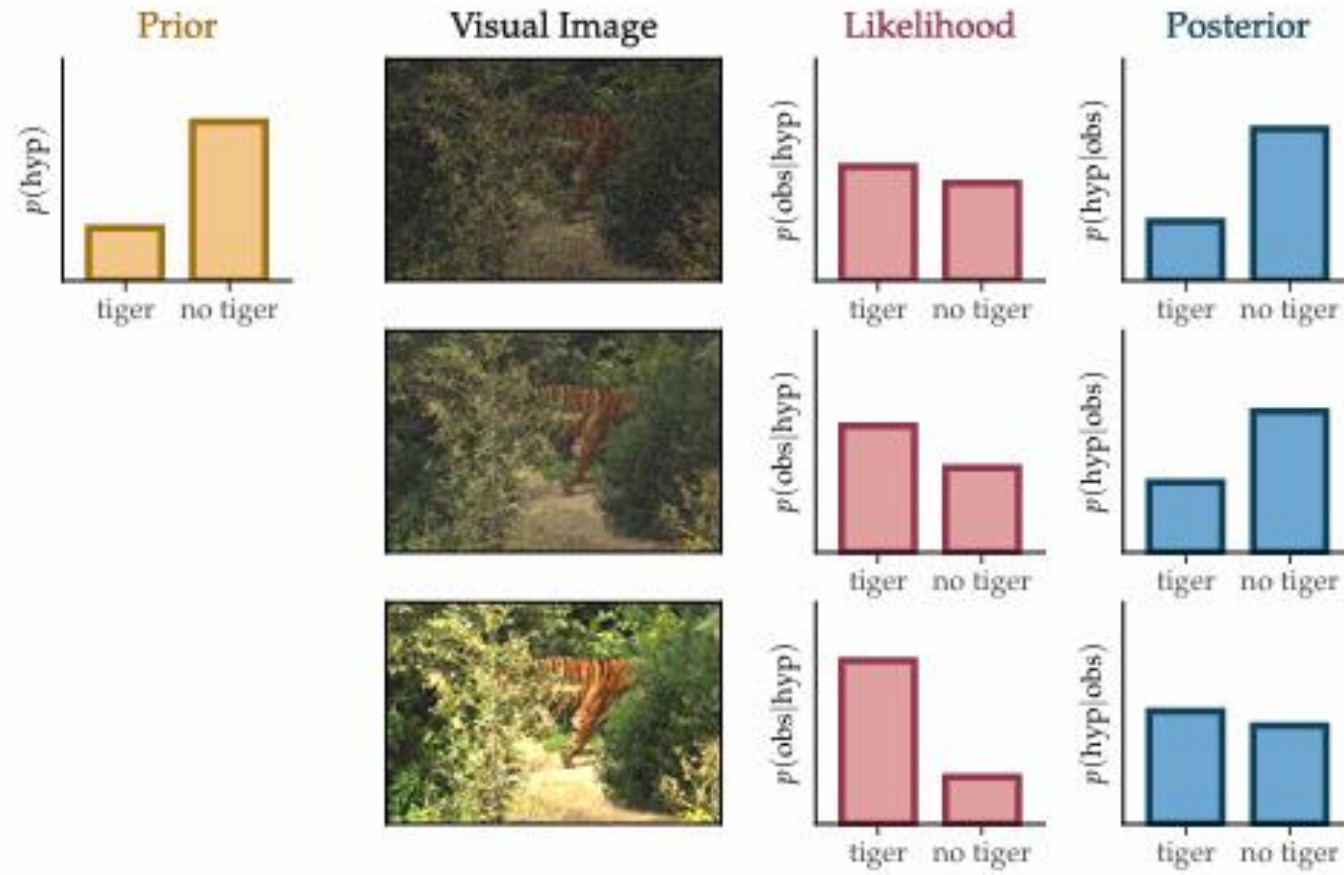


Figure 1.8: tiger detection with different acuity

Camouflage: an evolutionary view of likelihood flattening

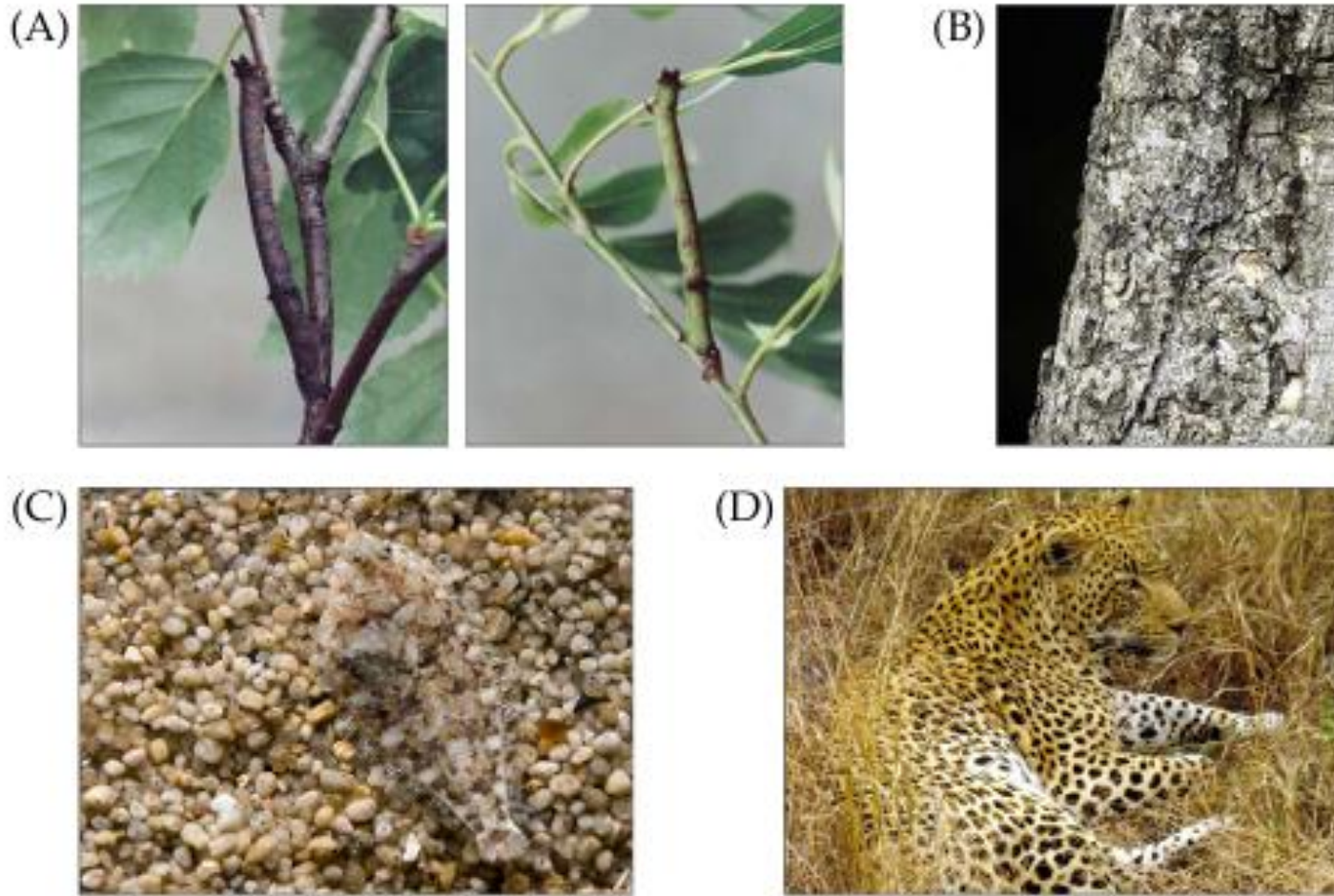


Figure 1.9: mimicry and blending widen likelihoods

1.5 Bayesian inference in auditory perception

Sound source localization: prior from vision + likelihood from hearing

Birds on a wire

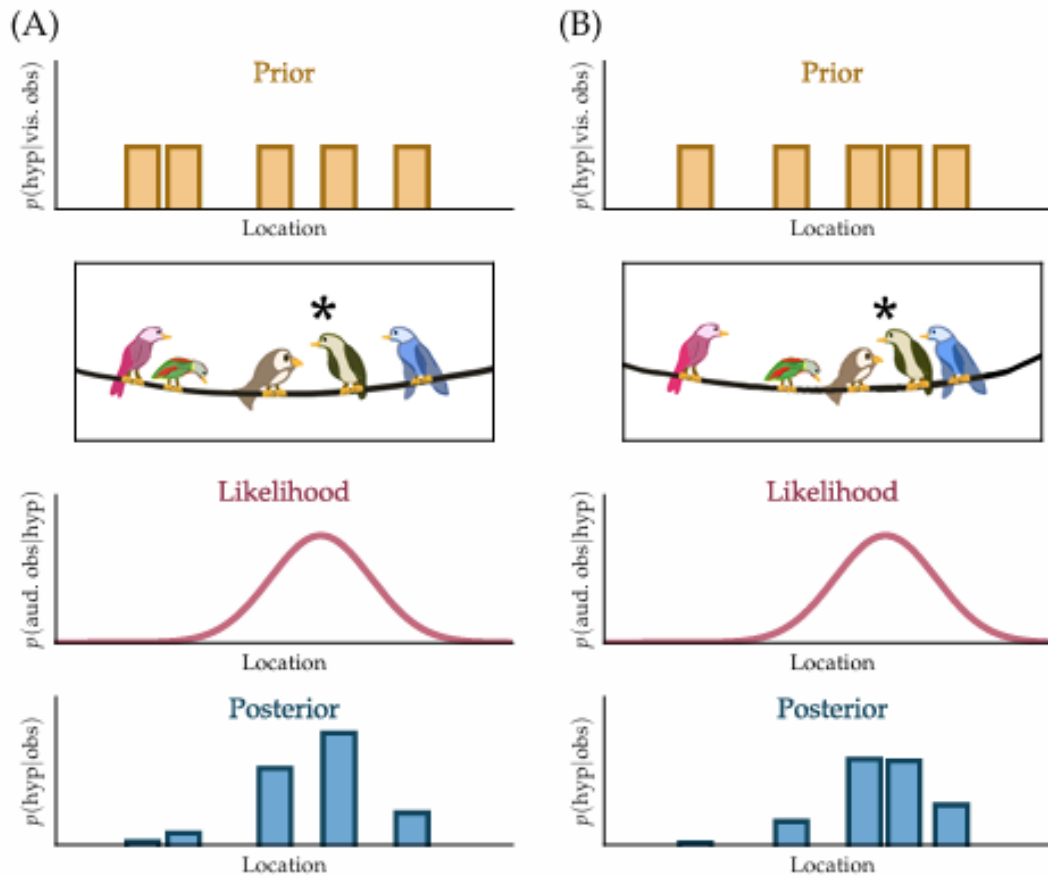


Figure 1.10: birds-on-a-wire example

Mondegreens: misheard words as Bayesian inference

Phonetic ambiguity (wide likelihood) + strong expectations (prior) → misperception

Lucy in the sky with diamonds
- The Beatles

"Lucy and this guy with diamonds"
"Lucy in disguise with diamonds"
"Lucy in the sky with Simon"



There's a bad moon on the rise
- Creedence Clearwater Revival
"There's a bathroom on the right"



The Death of Lady Mondegreen
- Harper's Magazine (Nov. 1954)

And you come to me on a summer breeze
- Bee Gees (How Deep Is Your Love)
"And you come to me on a submarine"

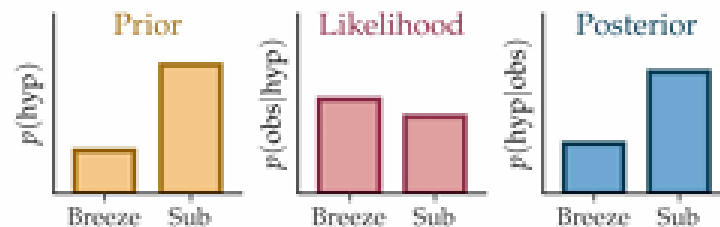


Figure 1.11: "Lucy in the sky..." examples

Semantic ambiguity: meaning-level uncertainty

Same words, multiple meanings → likelihood over meanings



Figure 1.12: Semantic ambiguity in language.

Syntactic ambiguity: structure-level uncertainty

Even with perfect hearing, you still infer “what was meant”.

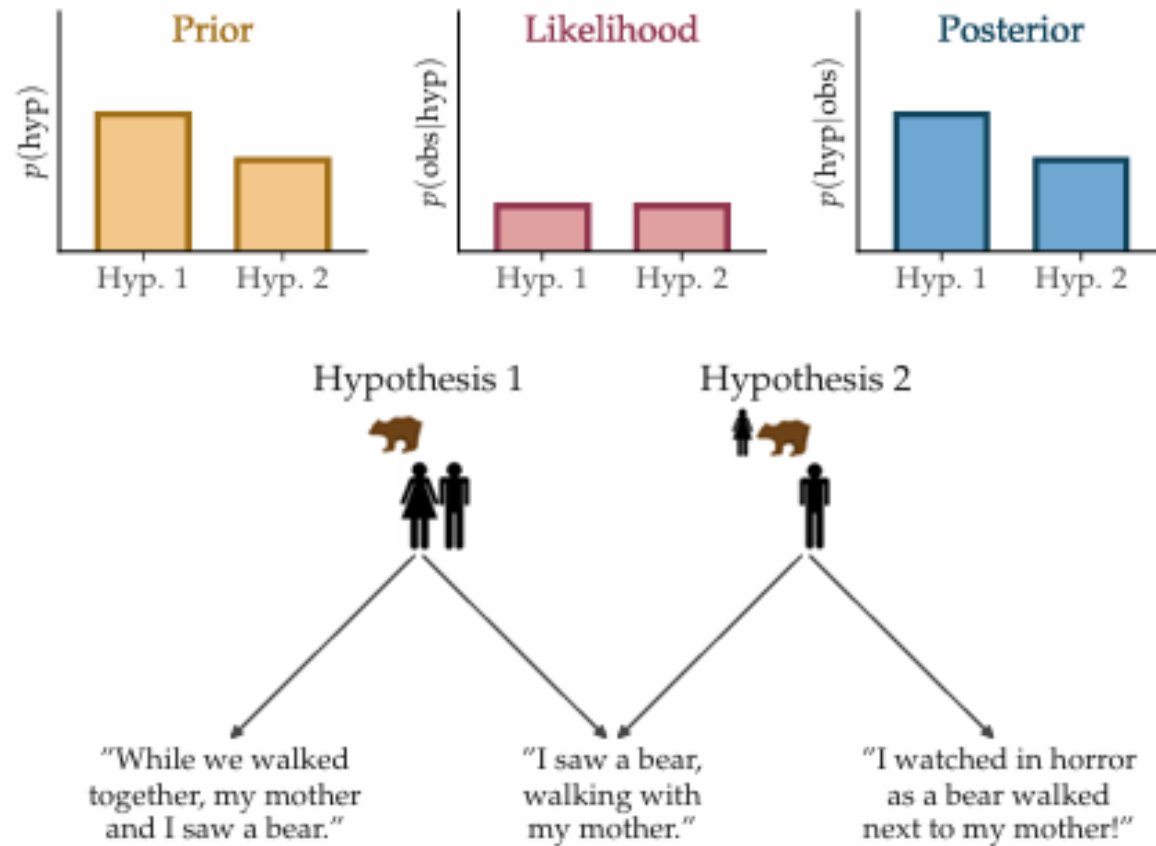


Figure 1.13: syntactic ambiguity + highlighted takeaway

1.6 Historical background: perception as unconscious inference



Thomas Bayes, 1702 — 1761



Ibn al-Haytham, 965 — 1040
"...familiar visible objects are perceived by sight through defining features and through previous knowledge..." — *De aspectibus*



Pierre-Simon Laplace, 1749 — 1827
"...the most important questions in life...are indeed, for the most part, only problems in probability. One may even say, strictly speaking, that almost all of our knowledge is only probable." — *Philosophical Essay on Probabilities*



Hermann Ludwig von Helmholtz, 1821 — 1894
"**Previous experiences** act in conjunction with **present sensations** to produce a **perceptual image**." — *Treatise on Physiological Optics*

$$\frac{p(\text{hyp}|\text{obs})}{\text{Posterior}} \propto \frac{p(\text{obs}|\text{hyp})}{\text{Likelihood}} \cdot \frac{p(\text{hyp})}{\text{Prior}}$$

Figure 1.14: Bayes, al-Haytham, Laplace, Helmholtz

1.7 Summary: the core pieces to remember

1.7 Summary

In this chapter, we have introduced the concept that perception is inherently probabilistic, and as such optimally characterized as a process of Bayesian inference. Regarding Bayesian inference, we have learned the following:

- Conditional probabilities such as $p(A|B)$ represents the probability of A given B . In Bayesian perceptual inference, A and B typically represent a world state and an observation.
- The likelihood function, $p(\text{observation}|\text{world state})$, captures the information content of the sensory observation, relevant to distinguishing one world state from another.
- The flatter the likelihood function, the less we learn from our senses. If the likelihood function is perfectly flat, then the observer has learned nothing from the observation.
- In some cases, such as the “Is that my friend?” example, the likelihood changes over time.
- The prior distribution over world states, $p(\text{world state})$, summarizes the information content of our past observations, the background knowledge we have about the world. Perception is not based entirely on sensory observation, but also on expectation grounded in previous experience.
- Flatter prior distributions mean we know less about the potential states of the world.
- Bayes’ rule calculates the posterior probability of each hypothesized world state, $p(\text{world state}|\text{observation})$, from the likelihoods and prior probabilities of the world states.
- The procedures of Bayesian inference apply equally to situations in which the hypothesized world states are discrete or in which they are continuous.
- Perceptual situations, whether in vision, audition, or other senses, are subject to various levels of uncertainty.
- Speech perception is fraught with phonetic and syntactic ambiguity, frequently giving rise to flat likelihood functions. The combination of priors and likelihoods can cause misinterpretations such as mondegreens.